5G MAG The Media Connectivity Association



OVERVIEW AND UPDATES · CONTENT DELIVERY AND STREAMING

Media Streaming and Advanced Media Delivery

Driving collaboration between mobile networks and media applications

3GPP CTA WAVE DASH-IF

NOVEMBER 2026 – www.5g-mag.com

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About the project

Driving collaboration between mobile networks and media applications

3GPP TS 26.501

Technical Specification

The functions, reference points and features of 5G Media Streaming, and the generalised Media Delivery architecture shared with RTC.

3GPP TS 26.512

Technical Specification

The protocols and APIs of 5G Media Streaming, specified with OpenAPI files.

3GPP TS 26.510

Technical Specification

The Media Delivery services, which harmonise media delivery across 5GMS and Real-Time media Communication.

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Features

The 13 features of 5G Media Streaming, by the release that added them

Release 16 the first version: eight features

- Content hosting
- Content publishing
- Content preparation
- Network assistance
- Dynamic policies
- Remote control
- Consumption reporting
- QoE metrics reporting

Release 17 edge, eMBMS and data collection

- Edge processing
- eMBMS delivery
- Data collection, reporting and exposure

Release 18 Service URL and MBS

- Service URL handling
- MBS delivery

Each feature applies to downlink, uplink or both: content hosting, consumption reporting, QoE metrics reporting and delivery via eMBMS and MBS are downlink; content publishing and remote control are uplink. None was added in Release 19.

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How it grew

What each 3GPP release added to 5G Media Streaming

Release 16

The first version of the architecture, procedures and APIs.

Release 17

Edge extensions and AF event exposure, and delivery via eMBMS.

Release 18

Service URL handling and delivery via MBS. TS 26.510 is created and media delivery is harmonised with RTC.

Release 19

Advanced Media Delivery: CMCD, multi-access, several service locations, encrypted content, QoS, network slicing.

Release 20

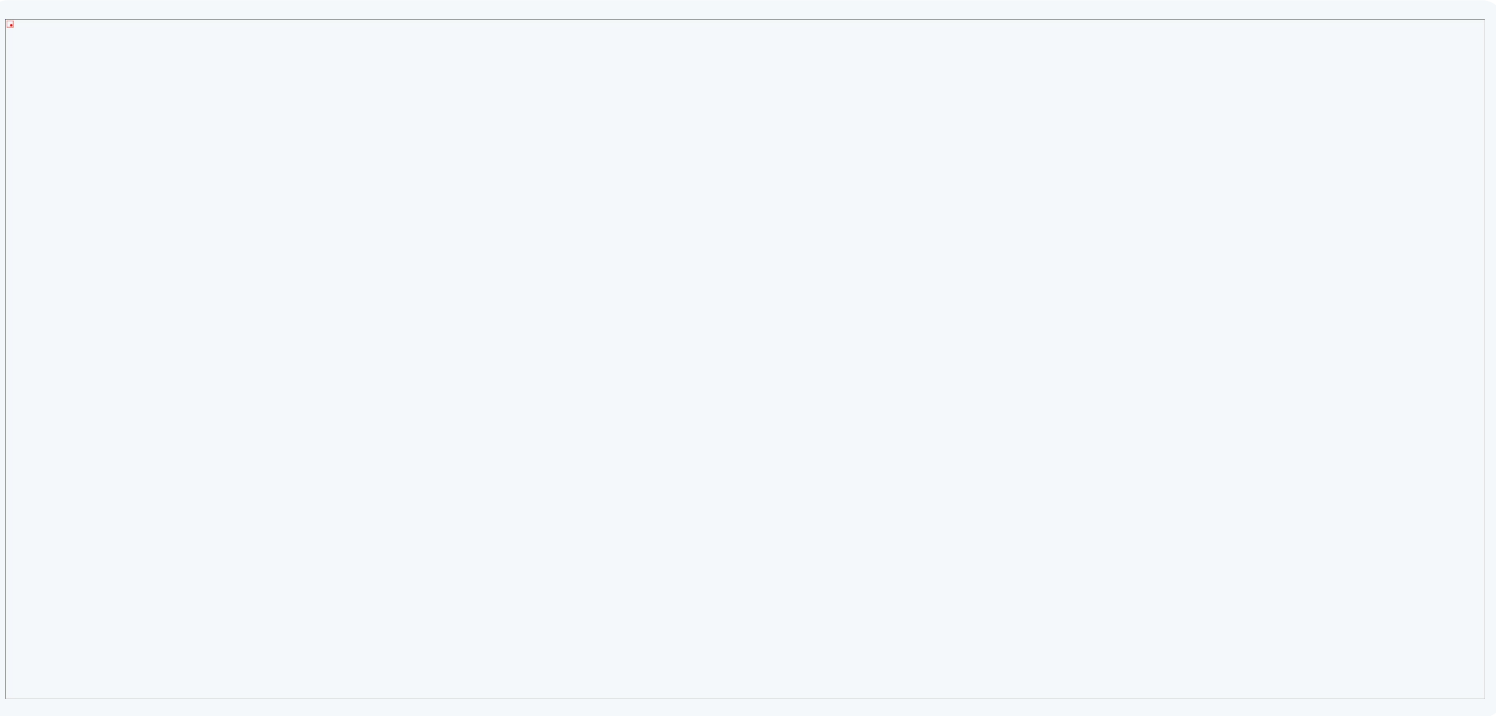
Advanced Media Delivery Phase 2: a stage 2 work item due December 2026 (work plan of 28 September 2026). Unverified beyond the TS 26.501 change history.

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Architecture

The 5GMS functions and the reference points between them



5GMS Application Provider

Uses 5GMS for its streaming services, and provides a 5GMS-Aware Application on the device.

5GMS AF

An Application Function dedicated to 5G Media Streaming, with control functions for the Media Session Handler and the Application Provider.

5GMS AS

An Application Server dedicated to 5G Media Streaming.

5GMS Client

In the device: a Media Session Handler, and a Media Player (downlink) or a Media Streamer (uplink).

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Repositories: network and tools

The network side of 5G Media Streaming and its tools, from the descriptions of the repositories

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-5gms-application-function	the 5GMS Application Function (AF), a Network Function of 5G Media Streaming	Application Function	v1.4.7 3 Aug 2026	GitHub
rt-5gms-application-server	the 5GMS Application Server (AS), a Network Function of 5G Media Streaming	Application Server	1.2.5 3 Aug 2026	GitHub
rt-5gms-application-provider	tools to work with the AF: command line, Management UI, Postman collection, QoE Metrics Reporting UI	Application Provider	1.3.0 3 Aug 2026	GitHub
rt-5gms-examples	a Docker Compose setup of 5G Media Streaming and an Express mock AF	Deployment	v1.3.0 3 Aug 2026	GitHub
cmcd-toolkit	tools to collect and analyze CMCD v2 data locally and across cloud providers	CMCD reporting	none yet	GitHub

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● released ● no release yet

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Repositories: the 5GMS Client

The client side, for Android, from the descriptions of the repositories

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-5gms-application	Android applications that test and demonstrate the Reference Tools: the 5GMSd-Aware Application and the Exo DVB-I Player	5GMS-Aware Application	v1.3.0 26 May 2026	GitHub
rt-5gms-media-session-handler	a 5GMS Client component for Android	Media Session Handler	v1.3.0 26 May 2026	GitHub
rt-5gms-media-stream-handler	a 5GMS Client component for Android, built on ExoPlayer	Media Stream Handler	v1.3.0 26 May 2026	GitHub
rt-5gms-common-android-library	models and helper classes shared by the Android client applications	5GMS Client	v1.3.0 26 May 2026	GitHub

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● released ● no release yet

More information at 5g-mag.com/tech/5gms

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Repositories: media and core

The media origin, the 5G Core consumers and the CMMF encoder, from the descriptions of the repositories

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-media-origin	a media origin server and packager, a JavaScript and TypeScript web application, for delivery testing	Media origin	none yet	GitHub
rt-5gc-service-consumers	reusable service consumer libraries that use the 5G Core network functions through their service interfaces	Service consumers	2.1.1 5 Jun 2026	GitHub
rt-cmmf-encoder	a reference implementation of a CMMF encoder, according to ETSI TS 103 973	No encoder code yet	none yet	GitHub

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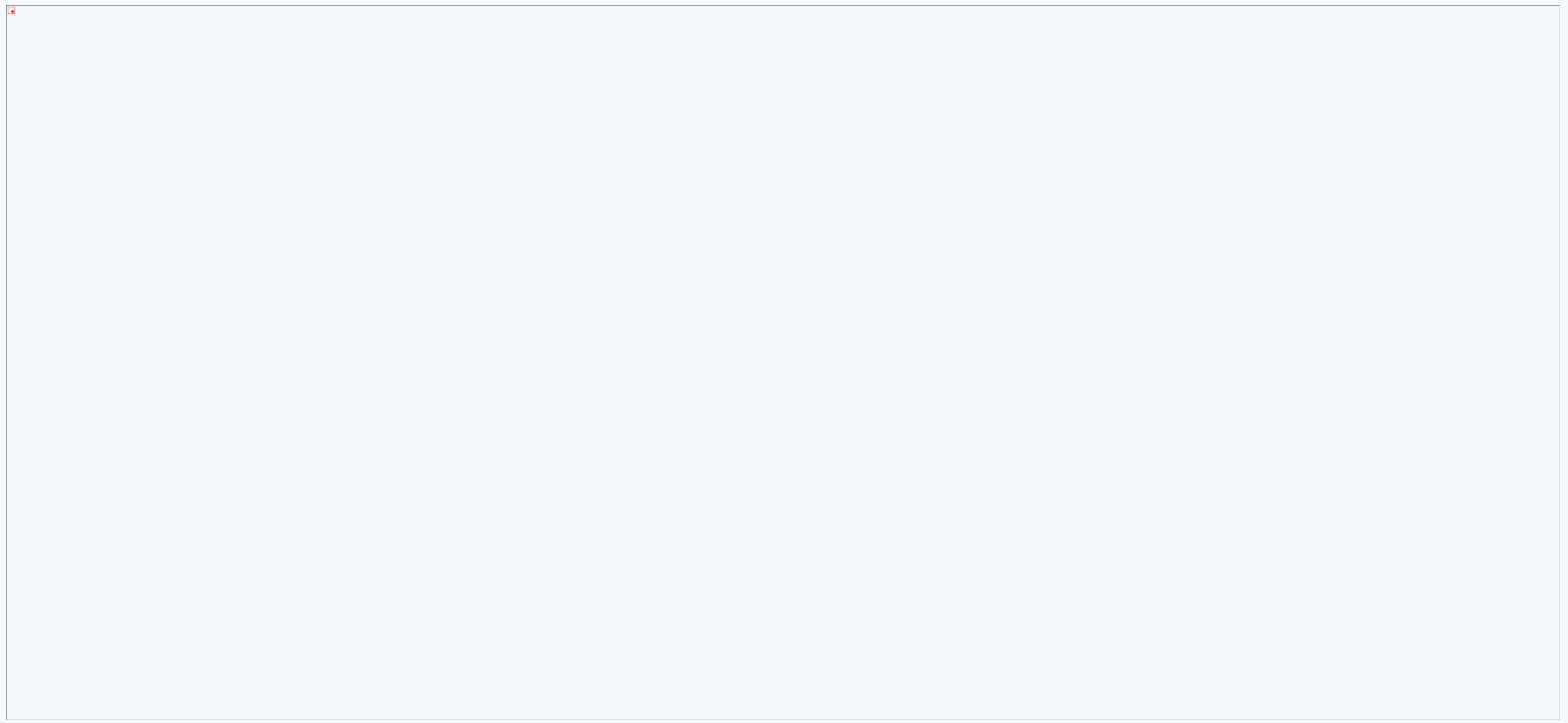
● released ● no release yet

More information at 5g-mag.com/tech/5gms

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Repositories in the architecture

The Reference Tools on the 5GMS architecture, as the figure generator draws them



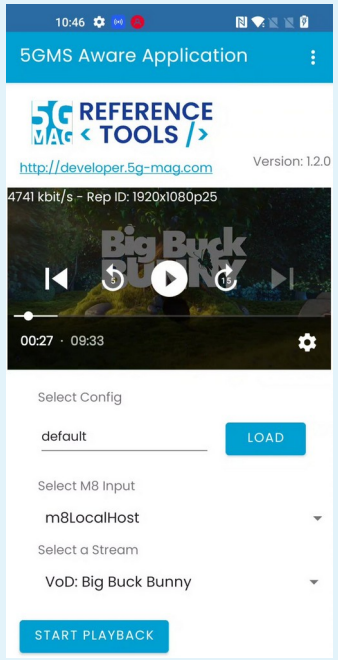
PROVIDED BY THE REFERENCE TOOLS

1. **rt-5gms-application**
2. **rt-5gms-application-provider**
3. **rt-5gms-application-function**
4. **rt-5gms-media-session-handler**
5. **rt-5gms-application-server**
6. **rt-5gms-media-stream-handler**
   used by 1
7. **rt-5gms-common-android-library**
   dependency, used by 1, 4, 6
8. **rt-common-shared**
   auxiliary library, used by 3, 5
9. **rt-5gc-service-consumers**
   dependency, used by 3
10. **open5gs**
    dependency, used by 3

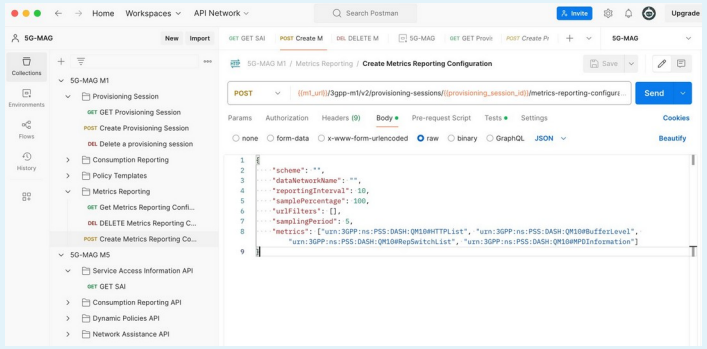
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See it and try it

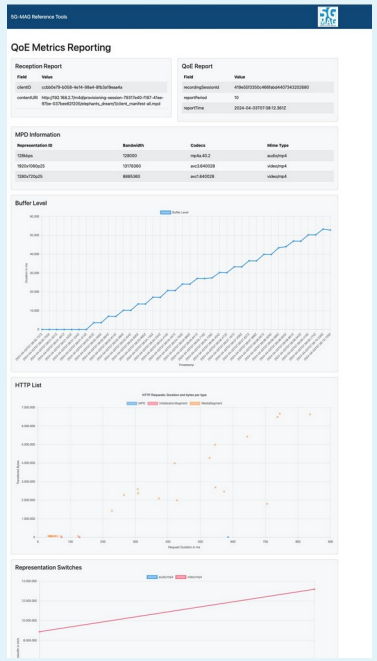
The 5GMSd-Aware Application, the provisioning of a streaming session, and the QoE reports it produces



5GMSd-Aware Application



Provisioning with the Postman collection



QoE Metrics Reporting UI

CONTRIBUTORS

BBC Fraunhofer FOKUS Dolby Qualcomm

WHERE WE STAND

5G Media Streaming (5GMS) Architecture

BROUGHT TO STUDY

- Study of implementation of 5GMS and RTC architectures
- Candidate topics from 3GPP AMD Phase 2

ON-GOING

- Implementation of Dynamic QoS policies
- Implementation of Network Assistance
- Implementation of CMMF support in 5GMS

OUTCOMES

- Implementation of Content Hosting
- Implementation of QoE Metrics and Consumption Reporting
- Initial Implementation of CMCD

RELATION WITH SDOS AND INDUSTRY FORA

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3GPP Feedback on TS 26.501, 26.510, 26.511 and 26.512: 15 issues open in the standards tracker, 8 raised since 23 June, 3 adopted

3GPP PCG Update presented on 21 October (PCG57_25)

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